

WEEK 3 REPORT

Financial Data Import, Transformation & Time Intelligence

(Power BI Desktop)

Submitted by
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1. Introduction

This week I used the Shopify stock data in Power BI Desktop to see how the share price and trading activity changed over the years. Below I have written the steps I followed, from cleaning the data to making the dashboard.

2. Objectives

- Import and prepare the stock data in Power Query.
- Create a Date table (Dim_Date) using CALENDARAUTO().
- Make a 1-to-many active relationship between Dim_Date and the stock data.
- Write DAX measures (Open, Close, High, Low, Trading Volume, Daily Return %).
- Use time intelligence functions: DATEADD, TOTALYTD, SAMEPERIODLASTYEAR.
- Create a Year-Quarter-Month-Day hierarchy.

3. Dataset

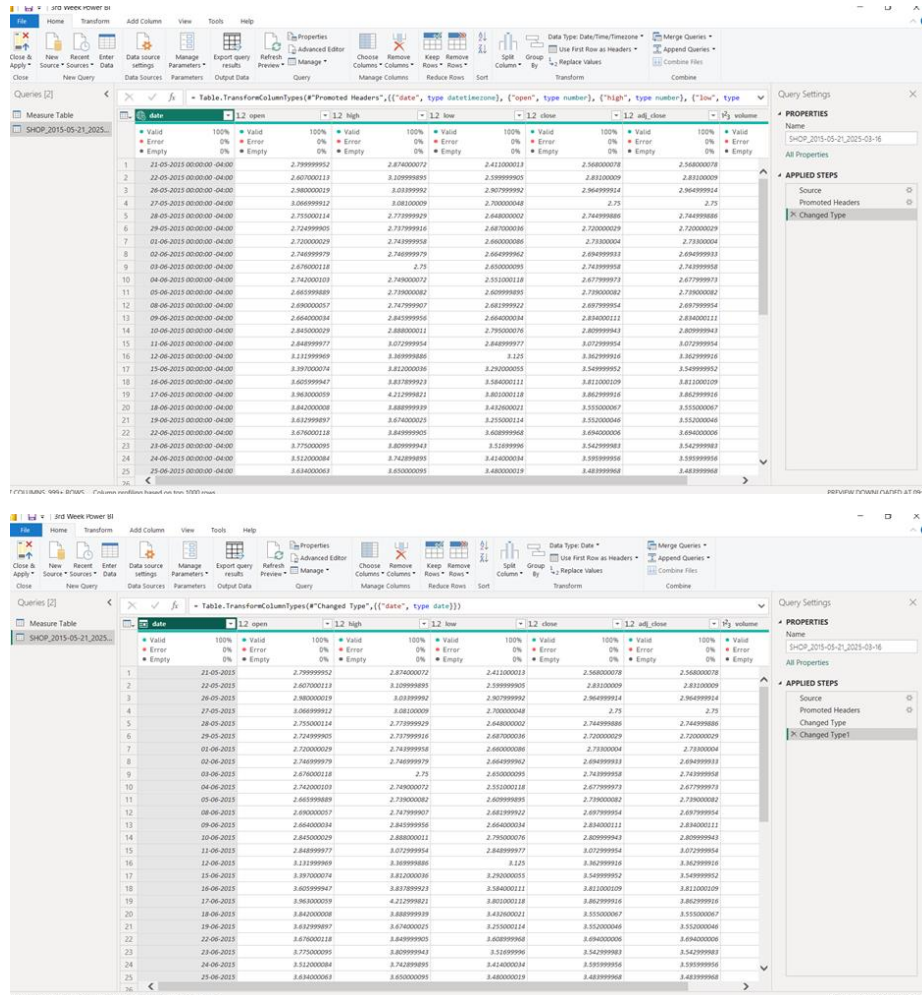
The data is daily Shopify stock data (SHOP_2015-05-21_2025-03-16). Each row is one trading day.

Column	Meaning
date	The specific trading date
open / close	Price at market open / close
high / low	Highest / lowest price of the day
adj_close	Close price adjusted for splits and dividends distributions of any stock
volume	The total Number of shares traded

4. Steps

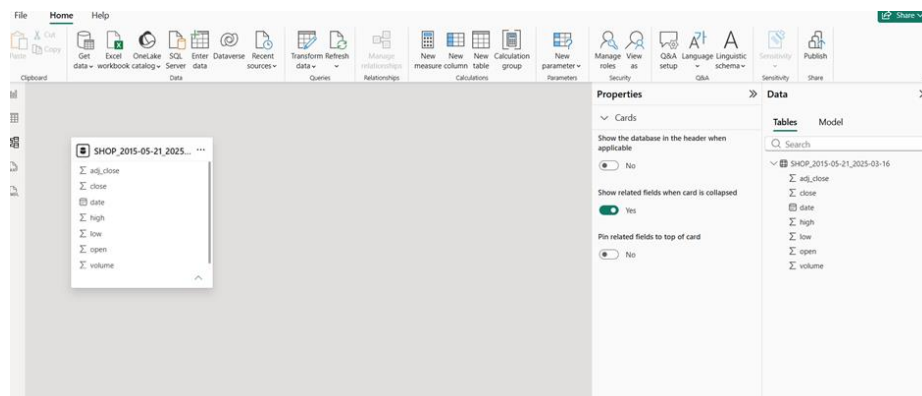
Step 1: Import and clean data in Power Query

I loaded the file in Power Query and used the first row as headers. The date column came with time and timezone, so I changed it to the Date type. All columns showed 0% errors and 0% empty, so no cleaning was needed. Then I clicked Close & Apply.



Step 2: Model view

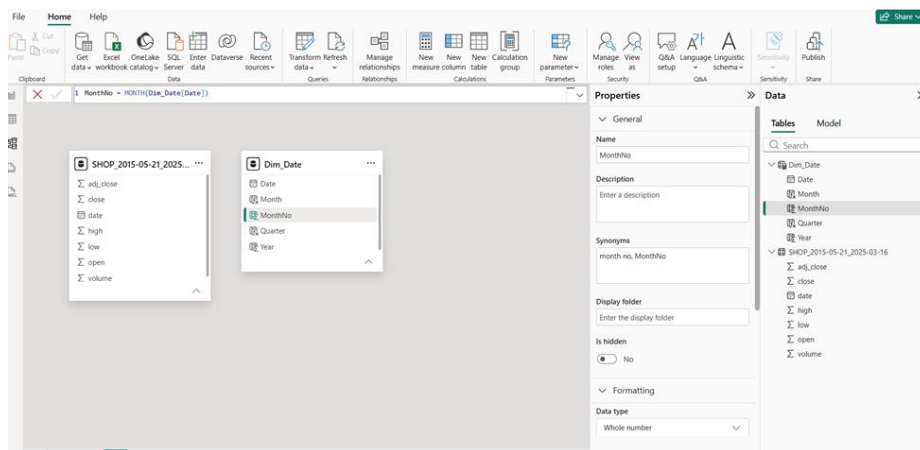
After loading, the table shows in Model view with all 7 columns.



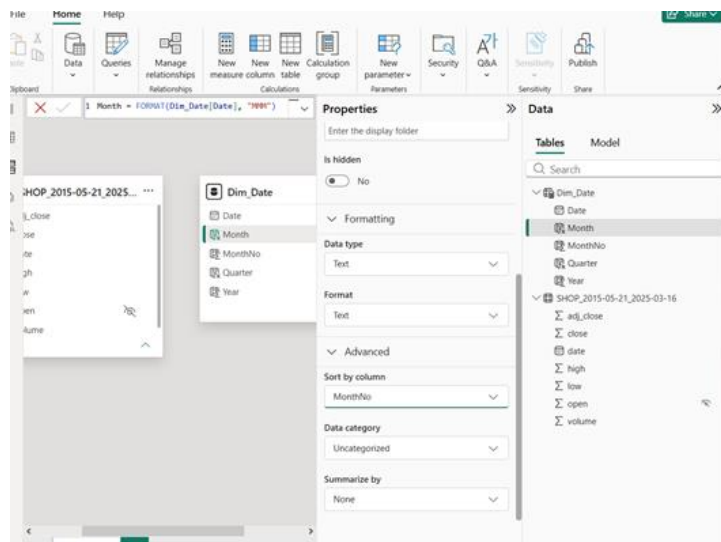
Step 3: Create Dim_Date table

I made the date table using CALENDARAUTO(). It creates one row for every day, starting from 01-01-2015. Then I added these columns:

```
Dim_Date = CALENDARAUTO()  
MonthNo = MONTH(Dim_Date[Date])  
Month = FORMAT(Dim_Date[Date], "MMM")  
Quarter = "Qtr " & QUARTER(Dim_Date[Date])  
Year = YEAR(Dim_Date[Date])
```

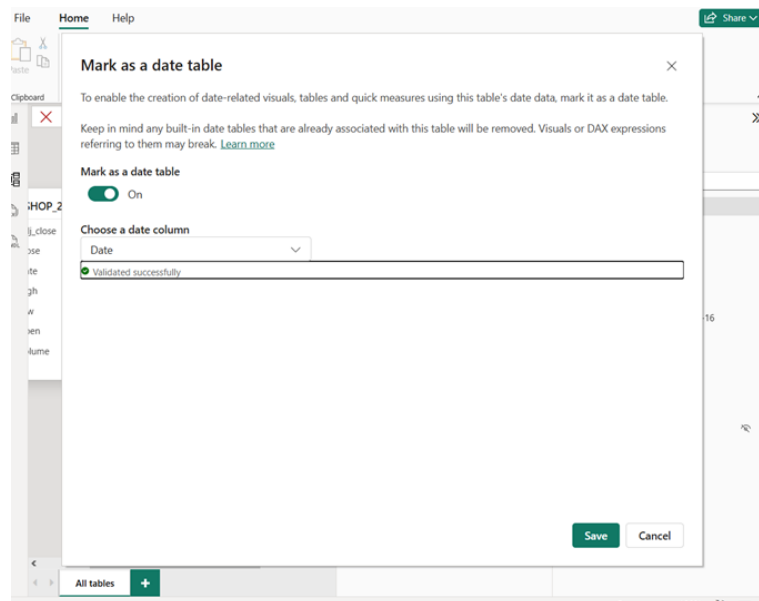


The Month column was sorting alphabetically, so I used **Sort by column** and selected MonthNo.



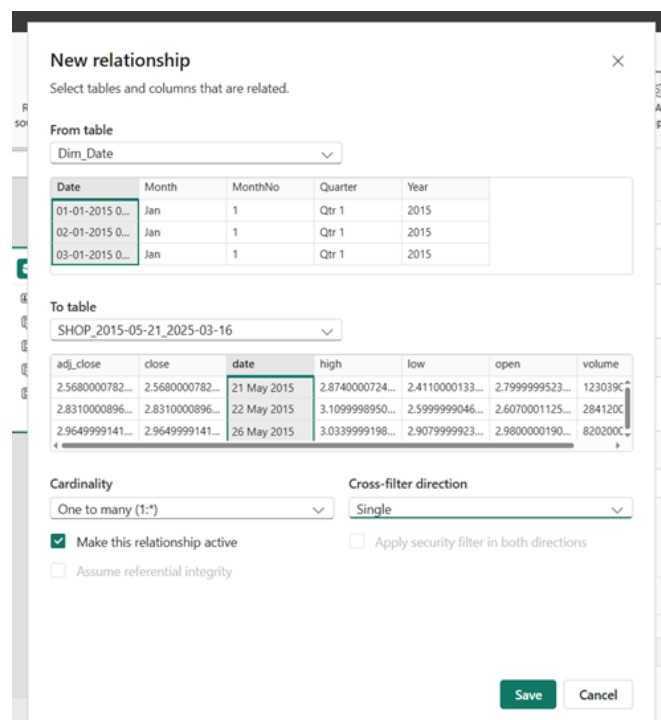
Step 4: Mark as date table

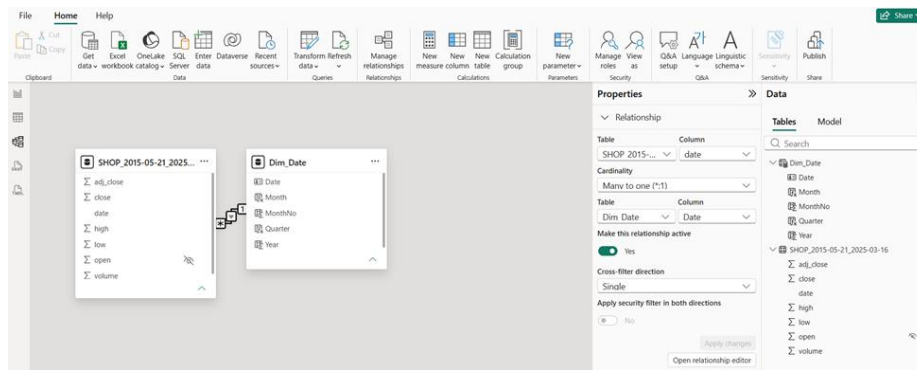
I selected Dim_Date, clicked Mark as date table and chose the Date column. It showed "Validated successfully".



Step 5: Create relationship

I connected Dim_Date[Date] to the stock table's date column with One to many (1:*), Single filter direction and Active ticked.



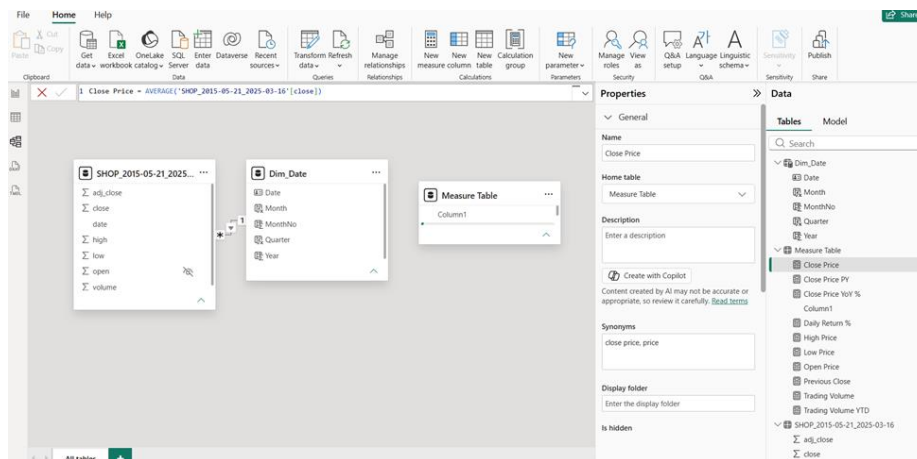


Step 6: Hierarchy

In Dim_Date I created a hierarchy with Year, Quarter, Month and Date, so I can drill down from year to day.

Step 7: DAX measures

I made a Measure Table to keep all the measures together.



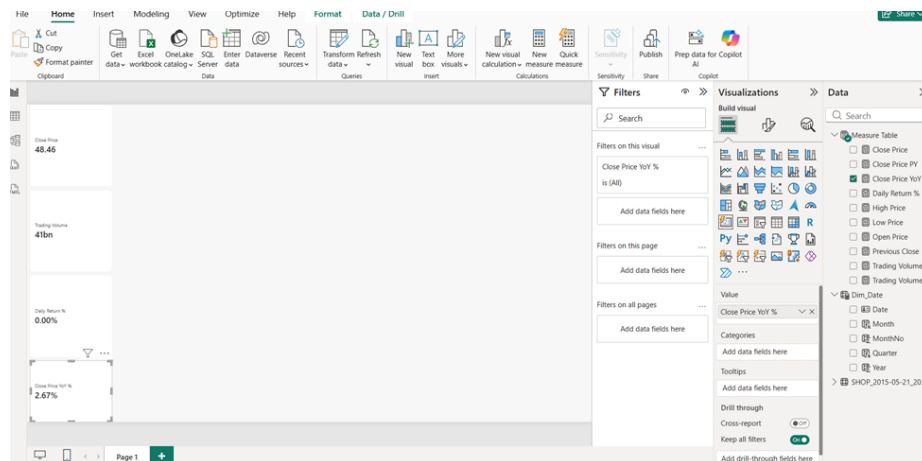
Measure	DAX formula	Meaning
Open Price	AVERAGE ('SHOP_2015-05-21_2025-03-16'[open])	Average open price
High Price	AVERAGE ('SHOP_2015-05-21_2025-03-16'[high])	Average high price
Low Price	AVERAGE ('SHOP_2015-05-21_2025-03-16'[low])	Average low price
Close Price	AVERAGE ('SHOP_2015-05-21_2025-03-16'[close])	Average close price
Trading Volume	SUM ('SHOP_2015-05-21_2025-03-16'[volume])	Total shares traded
Previous Close	CALCULATE([Close Price], DATEADD(Dim_Date[Date], -1, DAY))	Previous day close
Daily Return %	DIVIDE([Close Price] - [Previous Close], [Previous Close])	Change from previous day
Close Price PY	CALCULATE([Close Price], SAMEPERIODLASTYEAR(Dim_Date[Date]))	Close price last year

Measure	DAX formula	Meaning
Close Price YoY %	<code>DIVIDE([Close Price] - [Close Price PY], [Close Price PY])</code>	Growth vs last year
Trading Volume YTD	<code>TOTALYTD([Trading Volume], Dim_Date[Date])</code>	Volume from 1 Jan till date

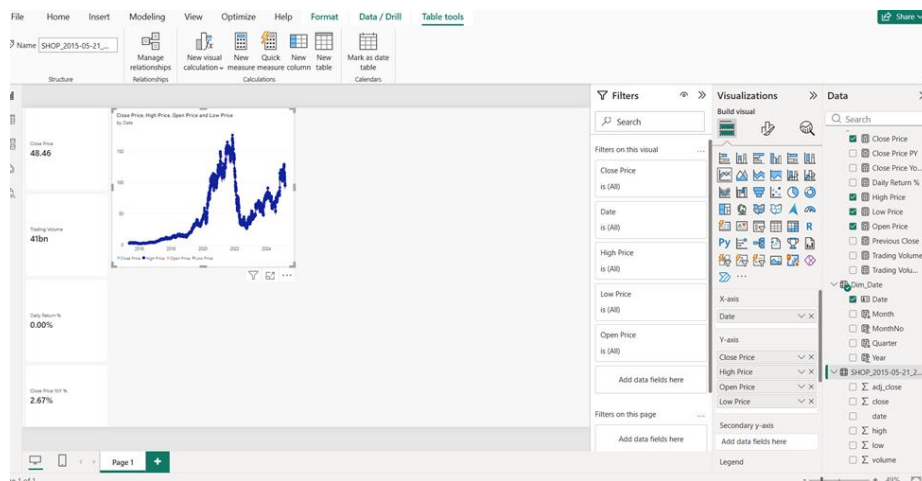
- **DATEADD** moves dates back or forward (used for previous day).
- **SAMEPERIODLASTYEAR** gives the same period of last year.
- **TOTALYTD** adds values from the start of the year till now.

Step 8: Visuals and dashboard

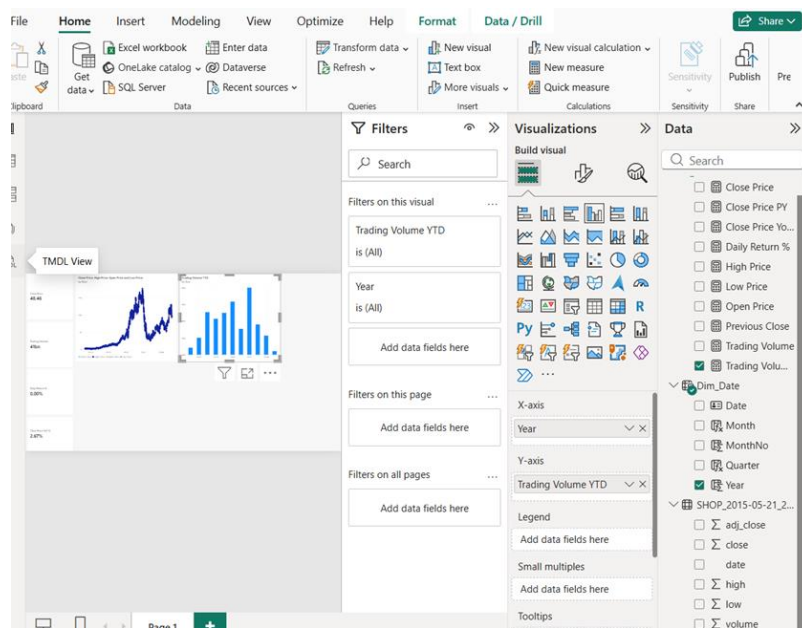
I made cards for Close Price, Trading Volume, Daily Return % and YoY %.



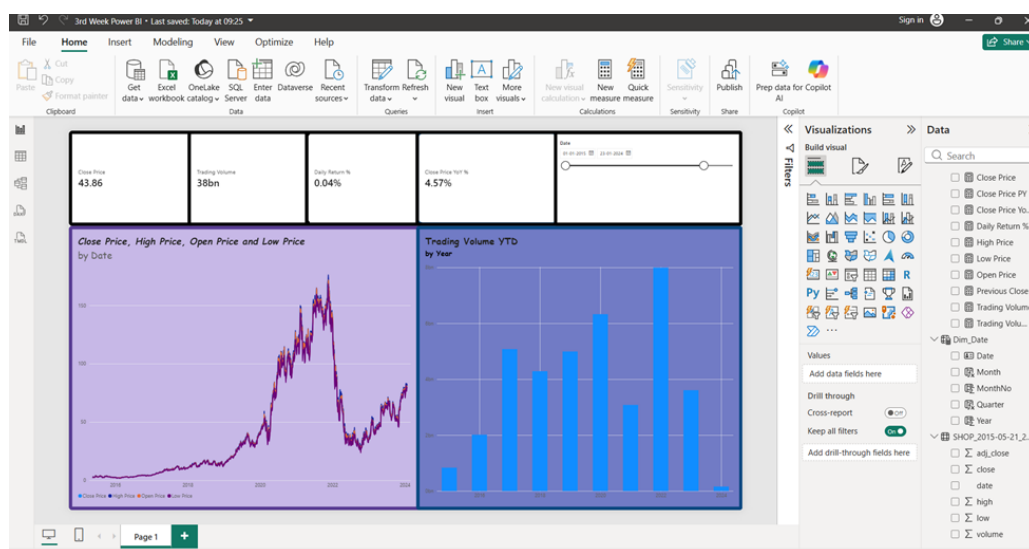
A line chart shows Close, High, Open and Low price by date.



A column chart shows Trading Volume YTD by Year.



Finally I added a date slicer and formatted the dashboard. The cards change when the date range changes.



6. Observations

- The price was low and grew slowly from 2015 to around 2019.
- It rose fast after 2020 and peaked around 2021.
- It fell sharply in 2022 and then started recovering.
- Trading volume was different every year.

7. Conclusion

This week I learned how to clean data in Power Query, make a date table, connect it to the data and write DAX time intelligence measures. The final dashboard helps an investor see the price trend and trading activity of Shopify easily.